FEBRUARY 25 • 10-MIN TEAM PRESENTATION

Streaming Data & Causal Baselines

Predictive Multitask Learning for Streaming Code-Switching

THE RESEARCH QUESTION

At time t, can we predict a language switch at t+1—and the length of the next segment—using $x_1 \ldots x_t$ only?

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Objectives & Universality

- Prediction, not detection: at word t, predict whether word t+1 switches language, using only $x_1 \ldots x_t$.
- Two targets: y_sw (switch or not) and y_dur (length of the new-language span: 1–2 / 3–6 / 7+ words).
- All 15 SwitchLingua pairs are in the pipeline (14 X–English + German–French); Cantonese–English is downsampled to the size of the others.

UNIVERSALITY

Same pipeline and metric on every pair: per-pair Anticipatory F1 and the SD across pairs. The pairs span 7 scripts, 7 language families, SVO / SOV / VSO word order, and one pair without English.

Language Coverage & Measured Pairs

ALL 15 PAIRS

Approach	Language Pair	Segmentation	Language Identification
Script-based	Arabic–English	Whitespace	Unicode (Arabic / Latin)
	Hindi–English (Devanagari)	Whitespace	Unicode (Devanagari / Latin)
	Korean–English	Eojeol	Unicode (Hangul / Latin)
	Russian–English	Whitespace	Unicode (Cyrillic / Latin)
	Chinese–English	Jieba	Unicode (Han / Latin)
	Cantonese–English	PyCantonese	Unicode (Han / Latin)
	Japanese–English	GiNZA (Bunsetsu)	Unicode (Japanese / Latin)
Model-based	Hindi–English (Romanized)	Whitespace	MuRIL Hinglish LID
Frequency-based	Spanish–English	Whitespace	wordfreq (EN / ES)
	French–English	Whitespace	wordfreq (EN / FR)
	German–English	Whitespace	wordfreq (EN / DE)
	Italian–English	Whitespace	wordfreq (EN / IT)
	Malay–English	Whitespace	wordfreq (EN / MS + ID)
	Filipino–English	Whitespace	wordfreq (EN / FIL)
	Vietnamese–English	PyVi	wordfreq (EN / VI)
	German–French	Whitespace	wordfreq (DE / FR)

Streaming Pipeline

01 CSV rows

data_generation_result

→

02 Words + LID

segment per language, then word LID

→

03 Targets

- y_sw (Switch Label)
- y_dur (Duration Label)

→

04 Streaming

window $x_{t-31} \dots x_t$

On demand: StreamingCSDataset is a PyTorch IterableDataset whose generator reads one CSV row at a time and yields each example as it is needed (Korean: 347K examples streamed with a 1.1 MB memory peak).

No precomputed windows: each example carries window $x_{t-31} \dots x_t$, the LID of x_t from the prefix, y_sw and y_dur; optional XLM-R / mBERT ids, scored at the last subword of x_t.

Utterance end: the last word has no x_{t+1}, so it yields no example, and windows never cross into another utterance.

Fixed split: rows sharing a sentence of 3+ words are grouped, then split 70 / 10 / 20; no such sentence appears in two splits.

Sample item: window = [I, want, to] next word = 먹어요 y_sw = 1 y_dur = 1 (scored at subword ▁to)

Next-Token (Word-Level) Labels: Worked Example

"I want to 먹어요 정말 맛있는 김치찌개 today"

x_t x_t+1

Word-level labels, predicted at the last sub-word of each word.

AT PREFIX "I want to"

Current language: EN
Next word: 먹어요 (KO)
y_sw = 1
y_dur = 1 (KO span = 4 words, medium)

AT PREFIX "... 먹어요"

Current language: KO
Next word: 정말 (KO)
y_sw = 0 • y_dur = −1 (no switch)

Duration:
At a switch, y_dur = length of the new span: 0 = 1–2 words, 1 = 3–6 words, 2 = 7+ words;
non-switch positions y_dur = −1, ignored by the duration loss (ignore_index = −1).

Prefix-Only Causal Constraint

LEGAL INPUT

At position t: $h_t = f(x_1, \ldots, x_t)$. Labels may read the whole sentence; inputs must not include x_{t+1}, in training or testing.

CAUSAL TRANSFORMER • UPDATE 2, NOT BUILT YET

Planned: mask attention to positions $j > t$.
Switch and duration heads read h_t.

NAIVE BASELINES • IMPLEMENTED AND VERIFIED

Input = language pair + word x_t and its LID, from $x_1 \ldots x_t$ only. Prefix-only re-run: LID identical in 14 pairs (Vietnamese 99.9%); word x_t identical in 12, over 99% in Cantonese, Japanese, Vietnamese.

Yuchen Huang

Naive Baselines & Anticipatory F1

REFERENCES

Always stay (never predict a switch): F1 = 0 on all 15 pairs.
Always switch (every word): F1 set by the switch rate alone.
Current LID: knows only the language of x_t.

NAIVE BASELINE • CURRENT WORD

P(switch | pair, word x_t) from train; threshold tuned per pair on validation.
x_t only, no punctuation. Gain = Naive − Always switch.

Pair	Always-switch	Current LID	Naive (word)	Gain	Pair	Always-switch	Current LID	Naive (word)	Gain
Korean	.194	.250	**.675**	+.481	Hindi	.410	.410	**.624**	+.213
Japanese	.192	.283	**.524**	+.332	Spanish	.143	.162	**.353**	+.210
German	.140	.172	**.417**	+.278	Filipino	.308	.308	**.491**	+.183
German–French	.141	.175	**.417**	+.277	Vietnamese	.169	.175	**.349**	+.180
French	.132	.163	**.370**	+.238	Russian	.171	.200	**.348**	+.178
Chinese	.138	.167	**.369**	+.231	Cantonese	.187	.192	**.364**	+.177
Italian	.133	.162	**.347**	+.214	Malay	.212	.212	**.361**	+.149
Arabic	.166	.217	**.381**	+.214	Mean / SD (15)		.216	**.426**	SD .102

Scope: test split only (20% of each pair's rows, full data, fixed and deduplicated), 65K–110K word positions per pair; switch-class F1 on automatic labels.
Knowing the word itself, not just its language, roughly doubles F1 (mean .216 → .426). Korean gains most: words ending in -요 / -다 (sentence-final) are followed by a switch 97% / 87% of the time, versus 11% for other Korean words.

Zaixun Li

Challenges & Next Steps

- Subword pieces: models cut words into pieces, too fine for Korean (mBERT splits 72% of Korean words) and too coarse for Chinese (XLM-R joins 7.6% of words with the next one). We still score one answer per word, at its last piece.

- What counts as a word: it decides how many answers are scored. Cutting Japanese into small units adds many easy “no switch” words: current-LID F1 .13, versus .28 with word + particle units like Korean. Units must match across languages for a fair comparison.

- Mapping language IDs to words: 59.5% of Hindi rows are romanized (MuRIL Hinglish LID); Filipino and Malay word lists contain English loans; words switch inside themselves (presentation에서, nag-meeting).

- Next: hand-checked gold set per pair; segment the prefix step by step (the Chinese, Cantonese, Japanese and Vietnamese segmenters read the whole sentence and move up to 1.1% of word boundaries); stronger baselines; causal dual-head XLM-R / mBERT.

Labels are automatic (script, wordfreq, MuRIL) and spot-checked, not gold: these are honest baselines, not trained models.

Takeaways & Team Handoff

DEFINED

15-pair universality study; word-level labels; prefix-only model inputs.

IMPLEMENTED IN COMPANION CODE

Streaming loader for all 15 pairs, per-language segmentation + LID, fixed 70/10/20 split, naive baselines.

NEXT: VERIFY AND MODEL

Gold-label LID audit, causal dual-head transformer, cross-pair comparison with context models.

Questions?